

Ecommerce Sales Intelligence Dashboard

Project Objective

Analyze an e-commerce sales dataset using **Microsoft Excel, SQL, and Power BI** to identify sales trends, product performance, customer behavior, profitability, shipping efficiency, and regional insights. Develop an interactive 4-page dashboard that enables management to make data-driven business decisions.

Project Roadmap

Phase 1 – Business Understanding

- Step 1: Understand the business problem
- Step 2: Explore the dataset
- Step 3: Define KPIs
- Step 4: Define business questions

Phase 2 – Data Preparation

- Step 5: Data Audit
- Step 6: Data Cleaning (Excel)
- Step 7: Data Validation
- Step 8: Create Calculated Columns

Phase 3 – Excel Analysis

- Step 9: Pivot Tables
- Step 10: Charts
- Step 11: Initial Business Insights

Phase 4 – SQL Analysis

- Step 12: Database Setup
- Step 13: SQL Queries
- Step 14: Advanced Business Analysis

Phase 5 – Power BI

- Step 15: Data Modeling
- Step 16: DAX Measures
- Step 17: Dashboard Development
- Step 18: Interactive Visualizations
- Step 19: Business Storytelling

Phase 6 – Portfolio

- Step 20: Dashboard Polish

- Step 21: Business Insights Documentation
 - Step 22: GitHub Repository
 - Step 23: README Documentation
 - Step 24: Resume Project Description
 - Step 25: LinkedIn Project Showcase
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Project Workflow

Raw Dataset



Excel Data Cleaning



SQL Data Analysis



Power BI Dashboard



Business Insights



GitHub Portfolio



Resume & LinkedIn

Business Objective

Build a professional Business Intelligence solution that helps answer:

- How is the business performing?
 - Which products generate the highest revenue and profit?
 - Which customers contribute the most value?
 - Which regions perform best?
 - How efficient are shipping operations?
 - Where can profitability be improved?
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Project KPIs

Sales KPIs

- Total Sales
- Total Orders
- Average Order Value
- Monthly Sales Trend

Profitability KPIs

- Total Profit
- Profit Margin
- Loss-Making Products

Customer KPIs

- Total Customers
- Average Profit per Customer

- Sales by Segment
- Top Customers

Product KPIs

- Sales by Sub-Category
- Profit by Sub-Category
- Top 10 Products by Sales
- Top 10 Products by Profit
- Bottom 10 Loss-Making Products

Shipping KPIs

- Average Shipping Days
- Total Shipping Days
- Sales by State
- Sales by Region
- Profit by Region

Final Dashboard Structure

Page 1 – Executive Overview

Objective

Provide an executive summary of business performance.

Visuals

- Year / Month / Region / Segment Slicers
- Total Sales
- Total Profit
- Profit Margin
- Total Orders
- Total Customers
- Average Order Value
- Monthly Sales Trend
- Sales by Category
- Profit by Category
- Sales by Region
- Top 10 Products by Sales

Page 2 – Sales & Product Analysis

Objective

Analyze product performance and identify profitable as well as loss-making products.

Visuals

- Total Sales
- Total Profit
- Total Quantity
- Total Products
- Sales by Sub-Category

- Profit by Sub-Category
 - Top 10 Products by Sales
 - Top 10 Products by Profit
 - Bottom 10 Loss-Making Products
 - Product Performance Matrix
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Page 3 – Customer Analysis

Objective

Understand customer purchasing behavior and segment performance.

Visuals

- Total Customers
 - Average Order Value
 - Total Orders
 - Average Profit per Customer
 - Sales by Segment
 - Profit by Segment
 - Top 10 Customers by Sales
 - Top 10 Customers by Profit
 - Customer Details Matrix
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Page 4 – Shipping & Geographic Analysis

Objective

Evaluate shipping performance and regional business performance.

Visuals

- Total Shipping Days
 - Average Shipping Days
 - Total States
 - Total Cities
 - Sales by Region
 - Profit by Region
 - Sales by State (Bubble Map)
 - Average Shipping Days by Ship Mode
 - Shipping Details Matrix
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Business Questions

Sales

- What is the total sales?
 - How do sales change over time?
 - Which region contributes the highest revenue?
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Products

- Which sub-categories perform best?
- Which products generate the highest sales?

- Which products generate the highest profit?
 - Which products generate losses?
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Customers

- Which customer segment contributes the highest sales?
 - Who are the top customers?
 - What is the average order value?
 - Which customers generate the highest profit?
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Shipping

- Which shipping mode takes the longest delivery time?
 - Which regions perform best?
 - Which states contribute the highest sales?
 - How does shipping efficiency vary?
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Tools Used

- Microsoft Excel
 - MySQL
 - Power BI
 - DAX
 - Power Query
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Skills Demonstrated

- Data Cleaning
 - Data Validation
 - SQL Analysis
 - KPI Development
 - DAX Measures
 - Data Modeling
 - Interactive Dashboard Design
 - Business Intelligence
 - Business Storytelling
 - Data Visualization
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Dataset Information

- Total Records: **9,995**
- Total Columns: **21**
- Date Range: **2014–2017**
- Industry: **E-Commerce**